DS	**Session**	Graph	**Question Information**	Level 1 Challenge 86

Question description

There are n boys and m girls in a school. Next week a school dance will be organized. A dancing pair consists of a boy and a girl, and there are k potential pairs.

Your task is to find out the maximum number of dance pairs and show how this number can be achieved.

Constraints

- $1 \le n,m \le 500$
- $1 \le k \le 1000$
- $1 \le a \le n$
- $1 \le b \le m$

Input

The first input line has three integers n, m and k: the number of boys, girls, and potential pairs. The boys are numbered 1,2,…,n, and the girls are numbered 1,2,…,m.

After this, there are k lines describing the potential pairs. Each line has two integers a and b: boy a and girl b are willing to dance together.

Output

First print one integer *rr*: the maximum number of dance pairs. After this, print lines describing the pairs. You can print any valid solution.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3 2 4
1 1
1 2
2 1
3 1
```

EXPECTED OUTPUT

```
2
1 2
2 1
```

Test Case 2

INPUT (STDIN)

```
3 2 4
1 1
1 2
2 3
3 2
```

EXPECTED OUTPUT

```
3
1 1
2 3
3 2
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void link(int u,int v)
```

Test Case 2

KEYWORD

```
int bfs(int n)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
2
```

Test Case 2

TOKEN COUNT

```
600
```

Test Case 3

NLOC

```
84
```

Code

school_dance.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <stdbool.h>

#define MAXN 505
#define MAXK 1005

int match_b[MAXN];
int match_g[MAXN];
bool visited[MAXN];
int adj[MAXN][MAXN];
int deg[MAXN];

void link(int u, int v) {
    adj[u][deg[u]++] = v;
}

int bfs(int n) {
    return n;
}

bool dfs_match(int u) {
    for (int i = 0; i < deg[u]; i++) {
        int v = adj[u][i];
        if (!visited[v]) {
            visited[v] = true;
            if (match_g[v] == 0 || dfs_match(match_g[v])) {
                match_g[v] = u;
                match_b[u] = v;
                return true;
            }
        }
    }
    return false;
}

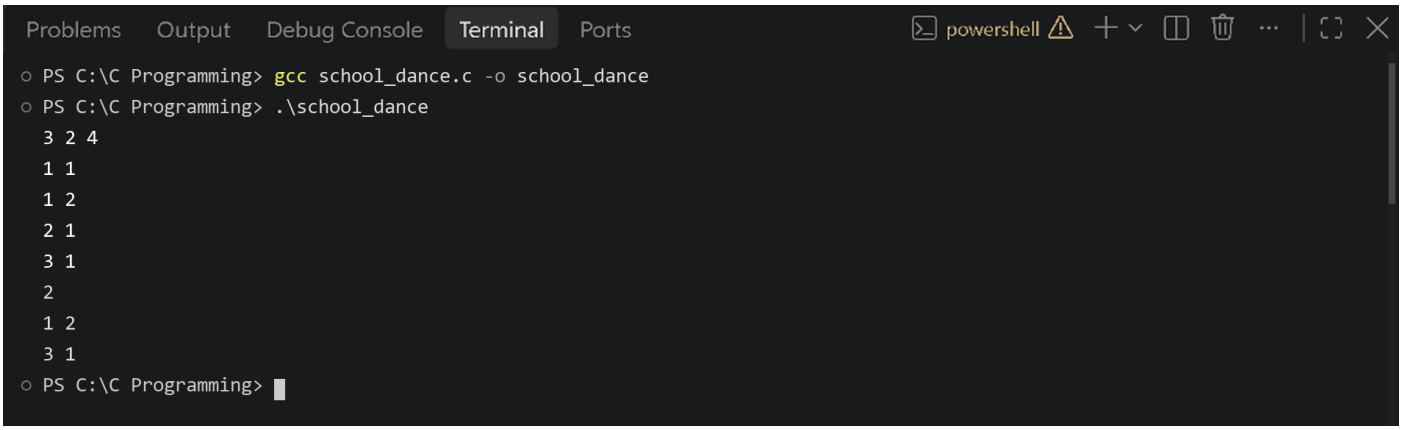
int main() {
    int n, m, k;
    if (scanf("%d %d %d", &n, &m, &k) != 3) return 0;

    for (int i = 0; i < k; i++) {
        int u, v;
        scanf("%d %d", &u, &v);
        link(u, v);
    }

    int total_pairs = 0;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) visited[j] = false;
        if (dfs_match(i)) total_pairs++;
    }

    printf("%d\n", total_pairs);
    for (int i = 1; i <= n; i++) {
        if (match_b[i] > 0) {
            printf("%d %d\n", i, match_b[i]);
        }
    }
    bfs(total_pairs);
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc school_dance.c -o school_dance
PS C:\C Programming> .\school_dance
3 2 4
1 1
1 2
2 1
3 1
2
1 2
3 1
PS C:\C Programming>